Free pythagorean theorem worksheet worksheet 8

Maths | 10th grade | Mixed Operations

1. If 5x - 3(2x + 4) = 2x + 7, what is the value of x?

A) 1
B) 3
C) 5
D) -1

2. Solve for y: 3(y + 2) - 2y = 2(y - 1) + 4

A) 0
B) -2
C) 2
D) 4

3. What is the value of the expression 7 + 3 * 2 - 8 / 4?

A) 11
B) 10
C) 9
D) 8

4. If 4(p + 5) = 2p + 18, what is the value of p?

A) 4
B) 3
C) 2
D) 1

5. Calculate the value of z if 9z - 4 = 5z + 12.

A) 4
B) 8
C) 6
D) 5

6. Simplify the expression: 2(x + 3) - 3(x - 2) + 4

A) x + 10
B) -x + 8
C) 5x + 4
D) -x + 12

7. If 5 - 2(a + 3) = a - 1, what is the value of a?

A) 1
B) -1
C) 2
D) 3

8. Find the value of w: 3w + 7 = 2(w - 1) + 9

A) 2
B) 5

C) 3
D) 4

9. Evaluate the expression: 6 + 2^3 * 2 - 5

A) 14
B) 15
C) 13
D) 16

10. If x/2 + 5 = 3x/4, what is x?

A) 10
B) 8
C) 12
D) 4

Answer Key

1. D
Expand and simplify: 5x - 6x - 12 = 2x + 7, leading to -x - 12 = 2x + 7, thus -3x = 19, x = -1.

2. A
Distribute and simplify: 3y + 6 - 2y = 2y - 2 + 4, yielding y + 6 = 2y + 2, leading to y = 4.

3. B
Follow order of operations: 7 + (3 * 2) - (8 / 4) = 7 + 6 - 2 = 11.

4. B
Expand and simplify: 4p + 20 = 2p + 18, leading to 2p = -2, thus p = 3.

5. A
Simplify to: 9z - 5z = 12 + 4, giving 4z = 16, thus z = 4.

6. D
Distribute and simplify: 2x + 6 - 3x + 6 + 4 = -x + 12.

7. A
Distribute: 5 - 2a - 6 = a - 1, simplifying to -2a - a = -1 - 11, a = 1.

8. D
Simplify: 3w + 7 = 2w - 2 + 9, hence w = 4.

9. D
Follow order of operations: 6 + (8 * 2) - 5 = 6 + 16 - 5 = 17.

10. B
Multiply throughout by 4: 2x + 20 = 3x, giving x = 8.